

Development Roadmap

10 phases: from critical fixes to college-scale optimization

Note on Excel vs MongoDB: Excel MongoDB ko replace nahi karega. MongoDB hi primary database rahega (users, resources, subjects, bookmarks, downloads). Excel sirf optional bulk import/export ya reporting ke liye use hoga (e.g. admin panel se resources ki list export karna, ya subjects/semesters bulk upload karna); data hamesha MongoDB mein store hoga.

Phase 1: Fix Critical Problems

- Create `userRoutes.js` and `userController.js`
- Add GET `/api/users/:id`
- Restore `/profile` and `/u/:userId`
- Fix Cloudinary `destroyAsset`
- Fix ResourceCard upvote state synchronization
- Fix AuthContext hydration race condition
- Fix existing critical bugs and broken functionality

Phase 2: Improve Architecture & Code Quality

- Add DELETE `/api/resources/:id`
- Add user authorization for resource deletion
- Delete resource from Cloudinary when deleted
- Add delete option for upload owners
- Add resource count aggregation to subjects
- Allow unauthenticated users to browse public resources
- Add `AbortController` to `useApi.js`
- Remove unused `App.css`
- Improve backend API structure and error handling

Phase 3: Authentication & User Features

- Add Logout
- Add Forgot Password
- Add secure password reset
- Add protected routes
- Add proper authentication state handling
- Add desktop user dropdown: Profile, My Semester, Saved Papers, Logout

Phase 4: UI/UX Improvements

- In-app PDF/document viewer
- Drag-and-drop upload
- File preview
- Upload validation
- Toast notifications
- Loading states
- Error states
- Empty states
- Responsive/mobile UI
- Proper title
- Favicon
- SEO meta tags

Phase 5: Notes, PYQ & Academic Resources

- Notes section
- PYQ section
- Syllabus
- Assignments
- Lab Manuals
- Question Banks
- Important Questions
- Study Material
- Other academic resources
- Search resources
- Filters: semester, subject, year, resource type, exam category

Phase 6: Admin Panel

- Admin login/access
- Admin dashboard: total users, total resources, total downloads
- User management: view users, view user uploads, manage users
- Resource management: add, edit, delete resources
- Upload academic files and replace files
- Manage Notes, PYQs, subjects, semesters, other academic content
- Backend-based admin authorization

Phase 7: Valuable Features

- Bookmark / Save Paper, Saved Papers tab, bookmark/unbookmark
- Download Counter
- Resource popularity
- Upvotes
- Categories: Mid-Sem, End-Sem, Class Test
- Academic year
- Tags

Phase 8: File Management

- Secure file upload
- PDF validation
- File size validation
- File type validation
- Binary magic-byte inspection
- Secure filenames
- Cloudinary integration
- Cloudinary file deletion
- Secure download/viewing

Phase 9: Production & Security

- `express-rate-limit`
- Helmet security headers
- Search sanitization
- ReDoS protection
- MongoDB text indexes
- MongoDB compound indexes
- Backend input validation
- Secure password hashing
- Secure password-reset tokens
- API authorization
- Admin authorization
- Environment variables
- No hardcoded secrets

Phase 10: College-Scale Optimization

- MongoDB as primary database
- Support approximately 500 college users
- Pagination
- Efficient MongoDB queries
- Optimized API calls
- Avoid unnecessary database requests
- Proper database indexes
- Cloudinary for files
- Efficient resource loading
- Search/filter optimization